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Pediatric Rheumatology

PPOM1

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Objectives

1. Basic concepts of Rheumatology
2. Common Pediatric Pathologies
 1. Juvenile Idiopathic Arthritis (JIA)
 2. Spondyloarthritides (Juvenile Ankylosing Spondyloarthritis, Psoriatic Arthritis, Reactive Arthritis)
 3. Pediatric SLE (pSLE)
 4. Neonatal Lupus
 5. Juvenile Dermatomyositis (JDM)
 6. Bechet disease (BD)
 7. Juvenile Sjogren's Disease (JSD)
 8. Pediatric Scleroderma
 9. FMF
3. Resources
4. Summarization



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Question:

3 yr old M brought into clinic by his mother for about 2 months of swelling in his knees. He went to the Urgent Care 3 months ago for a URI. It stops him from running around as much. He has also felt warm for the past 2 weeks especially in the evenings and some red/pink circles on his skin during that time which go away on its own.

On exam, his conjunctiva are pale bilaterally, there is palpable cervical, inguinal and popliteal lymphadenopathy, his liver is 3cm below the costal margin. His Vitals include HR 130 while calm.

If you got blood work on this patient, what do you think it would show ?



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Question:

3 yr old M brought into clinic by his mother for about 2 months of swelling in his b/l knees. He went to the Urgent Care 3 months ago for a URI. It stops him from running around as much. He has also felt warm for the past 2 weeks especially in the evenings and some red/pink circles on his skin during that time which go away on its own.

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Basic concepts

Rheum disease = History + Physical exam findings + autoimmune markers + serologic tests + pathology + imaging.

What clues you in that your patient may have a Rheumatologic condition?

Fevers

Arthralgias (joint pain) WITH arthritis (inflammation of the joint)

Timing (Morning stiffness – gets better as you use it more)

Rash

Weakness/Fatigue

Duration of symptoms

What you don't want to miss

Cancer (ALL/neuroblastoma)

Infection (viral illness, septic arthritis, lyme, osteomyelitis)

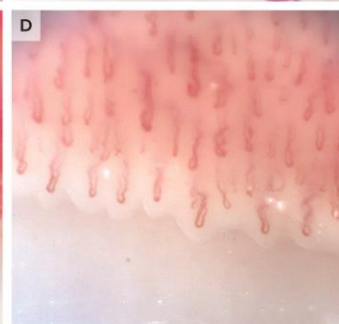
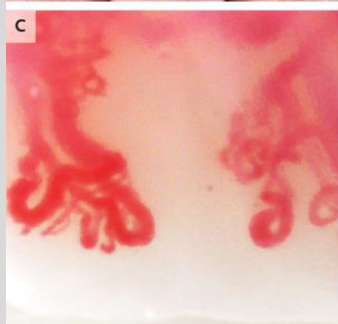
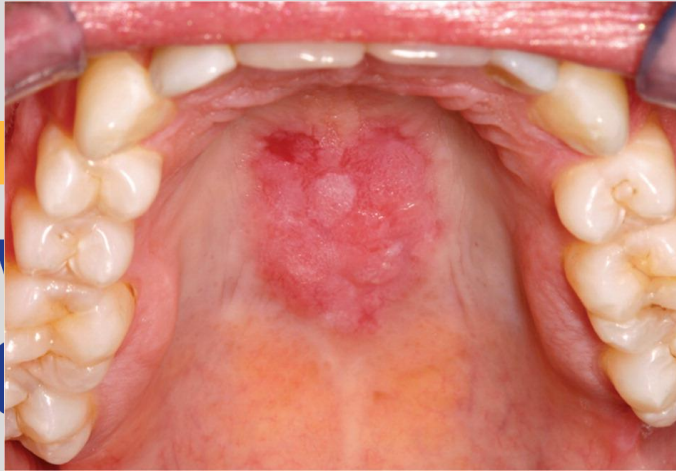
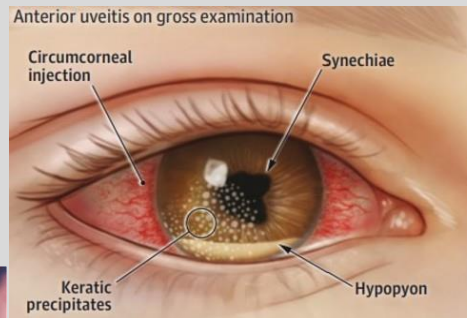
Mimics: metabolic (thyroid), orthopedic (patellofemoral syndrome, growing pains, chronic pain conditions (fibromyalgia, depression), other autoimmune disorders (IBD), vasculitidis

TABLE
Table 86.1

Differential Diagnosis of Pediatric Arthritis Syndromes

CHARACTERISTIC	SYSTEMIC LUPUS ERYTHEMATOSUS	JUVENILE IDIOPATHIC ARTHRITIS	RHEUMATIC FEVER	LYME DISEASE	LEUKEMIA	GONOCOCCEMIA	KAWASAKI DISEASE
Sex	F > M	Type-dependent	M = F	M = F	M = F	F > M	M = F
Age	10–20 yr	1–16 yr	5–15 yr	5–20 yr	2–10 yr	>12 yr	<5 yr
Arthralgia	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Morning stiffness	Yes	Yes	No	No	No	No	No
Rash	Butterfly; discoid	Salmon-pink macules (sJIA)	Erythema marginatum	Erythema migrans	No, petechiae	Palms/soles, papulopustules	Diffuse maculopapular, desquamation
Oligoarticular (up to four joints)	Yes	50%	No	Yes (often monoarticular)	Yes	Yes	—
Polyarticular	Yes	Yes	Yes	No	Yes	No	Yes
Small joints	Yes	Yes	No	Rare	Yes	No	Yes
Temporomandibular joint	No	Yes	No	Rare	No	No	No
Eye disease	Uveitis/retinitis	Uveitis/Iridocyclitis (rare sJIA)	No	Conjunctivitis, keratitis	No	No	Conjunctivitis, uveitis
Total WBC count	Decreased	Increased (decreased in MAS)	Normal to increased	Normal	Increased or neutropenia±blasts	Increased	Increased
ANA positive *	>99% positive	~50%	Nondiagnostic	Nondiagnostic	Nondiagnostic	Nondiagnostic	Nondiagnostic
Rheumatoid factor (RF) *	Positive or negative	Positive (10%) (polyarticular)	Nondiagnostic	Nondiagnostic	Nondiagnostic	Nondiagnostic	Nondiagnostic
Other laboratory results	↓ Complement, ↑antibodies to double-stranded DNA	Anti-CCP antibody + in adult type RA	↑ASO anti-DNase B	↑Cryoglobulin, ↑immune complexes	+ Bone marrow blasts	+ Culture for <i>Neisseria gonorrhoeae</i>	Thrombocytosis, ↑ immune complexes
Erosive arthritis	Rare	Yes	Rare	Rare	No	Yes	No
Other clinical manifestations	Proteinuria, serositis	Fever, serositis (sJIA)	Carditis, nodules, chorea	Carditis, neuropathy, meningitis	Thrombocytopenia	Sexual activity	Fever, lymphadenopathy, swollen hands/feet, mouth lesions

Basic concepts



For diagnosis: complete Physical Exam (as disease is multisystem)

Thorough Skin exam + scalp (generalized skin rash, sparing nasal labial folds, involving ears/around eyes, lower extremities, hair patterns)

Nail Beds (dilated capillary loops) - Nailfold capillaroscopy

Oropharynx: ulcers (hard palate involved in SLE), mucositis, TMJ joint instability/inflammation

Lymph nodes: lymphadenopathy

Eyes: uveitis, conjunctivitis

Heart: pericardial rubs

Lungs: pleural effusions, ILD

All Joints: Arthritis defined as intra-articular swelling of 2 or more of the following findings (pain on motion, range of motion limitation and warmth), limb length discrepancy



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Imaging: Plain xrays

MRI more sensitive for early
erosive arthritis

Echo

PFTs

CT chest

Basic labs:

Lymphopenia more specific to SLE than Leukopenia, in sJIA WBC Elevated
PLTs elevated – due to inflammation

Hbg low – anemia of chronic disease – nonspecific

ESR, CRP elevation – non specific

Rheumatoid Factor + is present in <10% of children with JIA – poor
sensitivity. Co-relate with symptoms = strong prognostic indicator

Anti- cyclic citrullinated protein (CCP) = RF pos JIA pretty specific

Anti- nuclear antibody – low specificity – not recommended for screening
but correlate with symptoms

DsDNA = SLE

Anti-Smith = SLE

SSA (Ro) positive = SLE, sjogren syndrome

SSA (La) positive = SLE, sjogren syndrome

Anti-histone = Drug induced lupus

Anti-neutrophil cytoplasmic antibodies (ANCA) = vasculitis

Anti- topoisomerase (Scl-70) = systemic sclerosis – rare in childhood

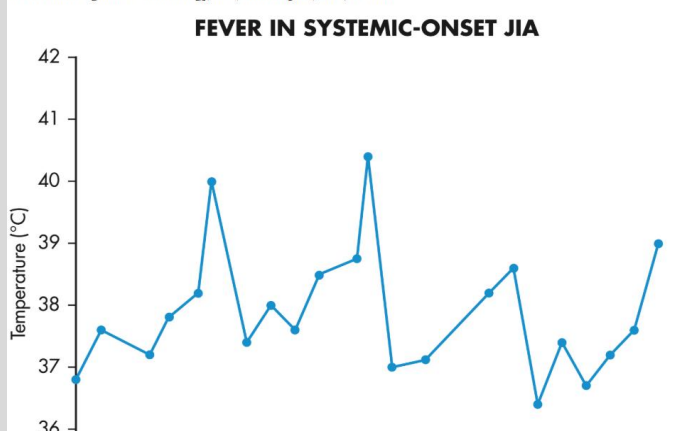
Muscle Enzymes – ALT, AST, CK, Aldolase, LDH

Urine – protein, can do protein/creatinine ratio

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From Hochberg MC: *Rheumatology*, ed 7, Philadelphia, 2018, Elsevier.



Common Pathologies

JIA (Juvenile Idiopathic Arthritis)



Most common rheumatic disease in children, clinical diagnosis

Age at onset: **< 16 years old**

Arthritis: **> or = 1 joint**

Duration: **> or = 6 weeks**

Type of articular involvement - many subtypes :

Poly: **> or = 5 joints**

RF neg or RF pos

Oligo: **< or = 4 joints**

Systemic-onset disease: arthritis with rash and quotidian fever

Psoriatic arthritis

Enthesitis-related arthritis

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FIG. 196.2

Synovial biopsy specimen from a 10-yr-old child with oligoarticular juvenile idiopathic arthritis. There is a dense infiltration of lymphocytes and plasma cells in the synovium.

Epidemiology

JIA: wide ranging numbers depending on environmental exposure, immunogenic susceptibility, variations in diagnostic criteria, lack of population based data.

Estimated 300,000 children in US have arthritis

Oligoarthritis being the most common, peak age at onset 2-4 years of age. Polyarthritis is bimodal (2-4 yrs, 10-14 yrs), sJIA (1-5 yrs)

No sex predominance for systemic JIA. Otherwise most other subtypes: girls > boys, except Enthesitis which is more common in boys.

Etiology:

Complex genetic trait – affecting both humeral and cellular immunity – MCH I and II, protein tyrosine phosphatase non receptor 22 (PTPN22), TNF- alpha, IL-6, IL-1 alpha. Double hit - **usually with external factor.** T cells recruited for non-self synovial antigens, cause inflammation cascade, including adding B cells and complement

ILAR Subtype	Peak Age of Onset (yr)	Female:Male	Arthritis Pattern	Extraarticular Features	Investigations	Therapy
Oligoarticular	1-3	4:1	Knees > ankles	Uveitis in ~20%	ANA positive in 60%; inflammatory markers are usually normal or only slightly ↑	NSAIDs and intraarticular steroids; MTX
Polyarticular, RF negative	1-3 and 9-14 (biphasic)	3:1	Symmetric or asymmetric; small and large joints; cervical spine; TMJ	Uveitis in ~10%	ANA positive in 40%; RF negative; may have elevated inflammatory markers and anemia of chronic disease	MTX and NSAIDs as first-line; often require TNF inhibitors or other biologics
Polyarticular, RF positive	9-11	9:1	Aggressive symmetric polyarthritis, especially of the small joints (wrists, hands)	Rheumatoid nodules in 10%	RF positive; CCP positive (60%); elevated inflammatory markers and anemia of chronic disease	MTX, TNF inhibitors, other biologics. Long-term remission off medication is unlikely; early aggressive therapy is warranted
Psoriatic	2-3 and mid-adolescence (biphasic)	2:1 in younger patients; 1:1 in older children	Asymmetric arthritis (oligoarthritis > polyarthritis) of small and medium-sized joints; sacroiliitis in 10%-30%	Psoriasis in 40%-66%; uveitis in 10%-15%; dactylitis and enthesitis common; comorbid obesity	ANA positive in 50%; inflammatory markers normal or mildly elevated	MTX, TNF inhibitors, and other biologics; NSAIDs and intraarticular steroids as needed can consider anti-IL-17 or anti-IL-12/23 therapy
Enthesitis-related arthritis	10-13	1:4	Predominantly asymmetric, lower limb arthritis; sometimes axial skeleton (but less than adult AS)	Enthesitis in 60%-80%; acute anterior uveitis; association with IBD	60%-90% HLA-B27 positive	NSAIDs and intraarticular steroids; consider sulfasalazine, MTX; TNF inhibitors if needed
Systemic	2-4	1:1	Polyarticular, often knees, wrists, and ankles; also fingers, neck, and hips	Daily fever; evanescent rash; pericarditis; pleuritis; hepatomegaly; splenomegaly; lymphadenopathy	Anemia; ↑WBC; ↑ESR; ↑CRP; ↑ferritin; ↑platelets (normal or ↓in MAS)	Standing NSAIDs; IL-1 antagonists; systemic steroids

ANA, Antinuclear antibody; AS, ankylosing spondylitis; CRP, C-reactive protein; ESR, erythrocyte sedimentation rate; IBD, inflammatory bowel disease; ILAR, International League of Associations for Rheumatology; IL-1, interleukin-1; JIA, juvenile idiopathic arthritis; MAS, macrophage activation syndrome; MTX, methotrexate; NSAID, nonsteroidal antiinflammatory drug; RF, rheumatoid factor; TMJ, temporomandibular joint; TNF, tumor necrosis factor; WBC, white blood cell count.



UTE



FIG. 196.9
CT scan of the temporomandibular joint of a child with juvenile idiopathic arthritis exhibiting destruction on the right.

Joints affected:

Oligo: larger joints

Polyarthritis: both upper, lower joints. Can also affect TMJs, cervical spine involvement and hip disease

Enthesitis – lower limb joints

Treatment:

NSAIDs (Naproxen, Ibuprofen, Meloxicam)

DMARDs (Methotrexate PO or SC weekly)

Biologics (Anti-TNF-alpha, anti-IL1 receptor)

Intra-articular corticosteroids

Systemic steroids – generally avoided unless bridging or severe acute disease



FIG. 196.11
Severe hip disease in 13-yr-old male with active systemic juvenile idiopathic arthritis. Radiograph shows destruction of the femoral head and acetabula, joint space narrowing, and subluxation of left hip. The child had received corticosteroids systemically for 9 years.



FIG. 7.7

Systemic juvenile idiopathic arthritis. (A and B) The rash is erythematous, macular, and often evanescent. It can be more prominent during periods of fever. Featured on trunk and extremities.

Prognosis:

Goal is disease remission

40-60% of patients get better with 6 weeks of NSAIDs

Oligo fares better than Poly

Depends on disease severity, earlier age at presentation, RF seropositive, disease involving hip, hand, wrist generally have poorer prognosis

Complications include leg discrepancy, micrognathia, contractures, persistent uncontrolled uveitis can cause chronic scarring and blindness

Governing bodies:
International League of Associations for Rheum Classifications (ILAR)
American College of Rheumatology (ACR)



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Table 196.3

Characteristics of ACR and ILAR Classifications of Childhood Chronic Arthritis

PARAMETER	ACR (1977)	ILAR (1997)
Term	Juvenile rheumatoid arthritis (JRA)	Juvenile idiopathic arthritis (JIA)
Minimum duration	≥6 wk	≥6 wk
Age at onset	<16 yr	<16 yr
Four or fewer joints in first 6 mo after presentation	Pauciarticular	Oligoarthritis: Persistent: four or fewer joints for course of disease Extended: four or more joints after 6 mo
Four or more joints in first 6 mo after presentation	Polyarticular	Polyarthritis, RF negative Polyarthritis, RF positive
Fever, rash, arthritis	Systemic onset	Systemic
Other categories included	Exclusion of other forms	Psoriatic arthritis Enthesitis-related arthritis Undifferentiated: Fits no other category Fits more than one category
Inclusion of psoriatic arthritis, inflammatory bowel disease, ankylosing spondylitis	No (see Chapter 197)	Yes

ACR, American College of Rheumatology; ILAR, International League of Associations for Rheumatology; RF, rheumatoid factor.

Anatomic region	Enthesitis exam
Foot and ankle	Achilles tendon insertion to calcaneus Plantar fascia insertion to calcaneus Plantar fascia insertion to metatarsal heads Plantar fascia insertion to base of 1st metatarsal
Knee	Quadriceps tendon insertion to patella (2 and 10 o'clock) Infrapatellar ligament insertion to patella (6 o'clock) and tibial tuberosity
Pelvis	Hip extensor insertion at greater trochanter of femur Sartorius insertion at anterior superior iliac spine Posterior superior iliac spine Abdominal muscle insertions to iliac crest Gracilis and adduction insertion to pubis symphysis Hamstring insertion to ischial tuberosity
Spine	5th lumbar spinous process
Upper extremity	Common flexor insertion at medial epicondyle of humerus Common extensor insertion at lateral epicondyle of humerus Supraspinatus insertion into greater tuberosity of humerus
Chest	Costovertebral junctions (1st and 7th)



FIG. 197.2

Anatomic sites for assessment of enthesitis in ERA and JAS.

Enthesitis related arthritis (ERA)

Classification Criteria

1. Arthritis PLUS Enthesitis
2. Arthritis OR Enthesitis with 2 of more of the following
 - Sacroiliac joint tenderness or inflammatory lumbosacral pain
 - Presence of HLA-B27
 - Onset of arthritis in a male older than 6 years
 - Acute anterior Uveitis
 - Family history of HLA-B27 associated disease in 1st degree relative
3. Typically first 6 months of disease are asymmetric and have fewer than 4 joints affected (inflammation of small joints in foot – highly suggestive of ERA)

Spondyloarthritides

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Clinical manifestations:

Arthritis of axial skeletal – spine or sacroiliac joint or hips
Enthesitis – inflammation at site of tendon, ligament or joint capsule
Eye inflammation – Acute Uveitis
GI inflammation

Overlapping Characteristics of the Spondyloarthritides *

From Petty RE, Laxer RM, Lindsley CB, et al., eds. *Textbook of Pediatric Rheumatology*, 8th ed. Philadelphia: Elsevier; 2021.

CHARACTERISTIC	JUVENILE ANKYLOSING SPONDYLITIS	JUVENILE PSORIATIC ARTHRITIS	INFLAMMATORY BOWEL DISEASE	REACTIVE ARTHRITIS
Enthesitis	+++	+	+	++
Axial arthritis	+++	++	++	+
Peripheral arthritis	+++	+++	+++	+++
HLA-B27 positive	+++	+	++	+++
Antinuclear antibody positive	–	++	–	–
Rheumatoid factor positive	–	–	–	–
SYSTEMIC DISEASE				
Eyes	+	+	+	+
Skin	–	+++	+	+
Mucous membranes	–	–	+	+
Gastrointestinal tract	–	–	++++	+++



FIG. 197.3
Loss of lumbodorsal spine mobility in a boy with ankylosing spondylitis. The lower spine remains straight when the patient bends forward.

Juvenile Ankylosing Spondylitis (JAS)

Frequently begins with oligoarthritis and enthesitis

Classification Criteria (modified New York criteria)

1. Arthritis PLUS Enthesitis
2. Arthritis OR Enthesitis with 2 of more of the following
 - Sacroiliac joint tenderness or inflammatory lumbosacral pain
 - Presence of HLA-B27 (90% of children with JAS)
 - Onset of arthritis in a male older than 6 years
 - Acute anterior Uveitis
 - Family history of HLA-B27 associated disease in 1st degree relative
3. Typically first 6 months of disease are asymmetric and have fewer than 4 joints affected
4. Treatment: goals of therapy to control inflammation, prevent ankylosis (fusion of bones) with DMARDS, NSAIDS, Physical Therapy
5. Prognosis: if ongoing disease activity for >5 yrs = disability very likely , remission in < 20% of those patients

Symptoms Characteristic of Inflammatory Back Pain

Pain at night with morning stiffness (and improvement on arising)

No improvement with rest

Improvement with exercise

Insidious onset

Good response to nonsteroidal antiinflammatory drugs

Psoriatic Arthritis

Classification Criteria

1. Arthritis PLUS Psoriasis
2. Arthritis OR Psoriasis with 2 of more of the following
 - Dactylitis
 - Nail pitting or onycholysis
 - Family history of Psoriasis in 1st degree relative
3. Disease onset peaks during preschool and early adolescent years

Rheumatology



FIG. 7.6

Psoriatic arthritis. (A) Erythematous plaques with silver scale. (B) Dactylitis (sausage toe) of the bilateral third and fifth toes. (C) Diffuse nail pitting. (D) Diffuse onycholysis.



FIG. 197.1

Nail pitting in a 9-year-old with JPSA.

Pamela F. Weiss, Chapter 197, 1483-1487.e1

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Reactive Arthritis

Quick review

Defined as joint inflammation caused by a sterile inflammatory reaction after recent infection 3d to 6wks

Can't see, can't pee, can't climb a tree – **not** common in kids. Usually more systemic symptoms (fatigue, weakness), cutaneous manifestations (erythematous oral macules, plaques, erosions, erythema nodosum, fingertip erosions and more enthesitis and dactylitis)

Arthritis is asymmetric, oligoarticular of lower extremities.

Bacteria: Salmonella, Shigella, Yersinia, Campylobacter, GU tract infections with Chlamydia, Ecoli and C diff less common

Virus: Adeno, CMV, EBV, HSV, Parvovirus, Rubella

75% of pts with reactive arthritis are HLA-B27 positive. Tx NSAIDS

If Chlamydia ongoing = TREAT the bug



FIG. 198.1

Enthesitis—swelling of the posterior aspect of the left heel and lateral aspect of the ankle.

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Mimics:

Acute Rheumatic Fever: caused by untreated GAS infection, 2-4 wks after illness. JONES criteria, will have migratory polyarthritis

Neisseria Gonococcal arthritis: rash, fever, polyarticular migratory, sexually active. Tx: CTX

Lyme Arthritis: can mimic oligoarticular JIA, arthritis can be a late manifestation months or years after lyme exposure, usually knee, can get titers to help with diagnosis

Pyogenic/Septic Arthritis: acute symptoms of inflammation of one joint, aspiration of joint fluid showed increased WBC with neutrophil predominance



Pediatric Onset SLE (pSLE)

Compared with adults, kids have more severe disease and widespread organ involvement

In adolescents: renal failure or nephrotic syndrome – 1st presenting sign

Prevalence of SLE in children 1-6/100,000 highest among African, Asian, Hispanic, Native American descent. Childhood SLE is rare before 5yr, median age at diagnosis 11-12 years. 20% of SLE pts are diagnosed before age 16yr.

Most common presenting sign in pSLE fever, fatigue, hematologic abnormalities, arthralgias and arthritis (symmetric polyarthritis large and small joints). Neuropsych complications hard on peds patients

Antiphospholipid antibodies found in 66% of pSLE

Tx: hydroxychloroquine, avoid sun exposure, steroids, MTX, biologics

Prognosis: 5yr and 10 yr survival with \$\$ countries 99, 97%, \$ countries 85, 79%



FIG. 7.13



FIG. 199.1

Mucocutaneous manifestations of SLE. A, Malar rash. B, Vasculitic rash on toes. C, Oral mucosal ulcers. D, Discoid rash in malar distribution.

Neonatal Lupus (NLE)

Not autoimmune disease but immunoglobulins Anti-Ro, Anti-La crossing over into fetal circulation

Moms are mostly asymptomatic, 50% have defined SLE or Sjogrens

Resulting in malar rash, congenital heart block (ranging from type 1 to type 3), cytopenias, hepatitis

It is thought there are Ro and La antigens on cardiac cells specially the AV node which then cause the fibrosis and varying degrees of block

Complete heart block is permanent – needs pacing, occasional IVIG, steroids used. Cardiomyopathy also a bad prognosis – need transplant

Data mixed but using steroids, IVIG to lower Anti-Ro and La antibodies seem to help prevent occurrence in future child (15-20%)

Prenatally, can detect heart abnormalities via Echo, helpful to see a ped cardiologist

Usual Treatment: time, avoidance of sun



FIG. 199.4

Neonatal lupus syndrome. Typical rash, often photosensitive with a malar distribution, appearing as annular plaques with erythema and scaling.



FIG. 200.3

The rash of juvenile dermatomyositis. The skin over the metacarpal and proximal interphalangeal joints may be hypertrophic and pale red (Gottron papules).

Diagnostic Criteria for Juvenile Dermatomyositis

Data from Bohan A, Peter JB. Polymyositis and dermatomyositis (second of two parts). *N Engl J Med*. 1975;292:403–407.

Classic rash	Heliotrope rash of the eyelids Gottron papules
Plus three of the following:	
Weakness	Symmetric Proximal
Muscle enzyme elevation (≥ 1)	Creatine kinase Aspartate transaminase Lactate dehydrogenase Aldolase
Electromyographic changes	Short, small polyphasic motor unit potentials Fibrillations Positive sharp waves Insertional irritability Bizarre, high-frequency repetitive discharges
Muscle biopsy	Necrosis Inflammation

Juvenile Dermatomyositis (JDM)

Characterized by proximal muscle weakness and distinctive rashes of face and extensor joint surfaces

3 cases in 1 million children in 1 year. Peak age onset 7 yrs. F:M 2:1

Genetics in HLA genes and TNF alpha, IL-1 receptor mutations with environmental factors

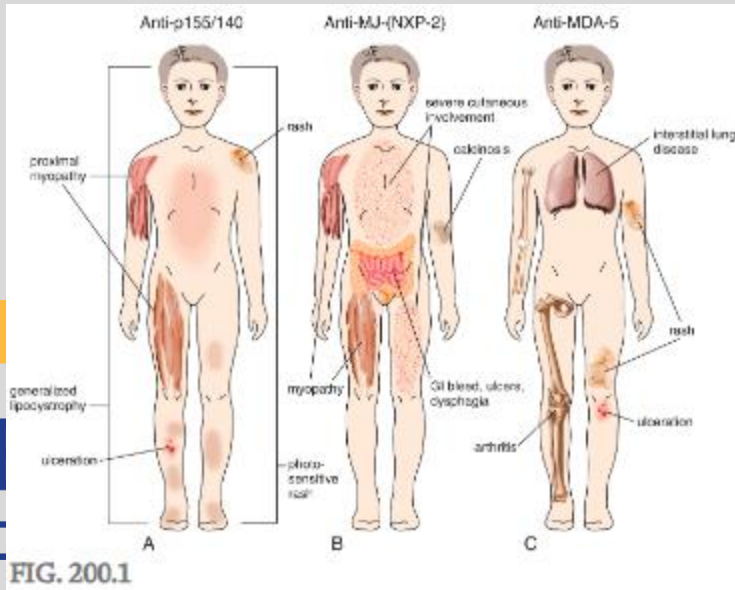
Children with JDM present with either rash, insidious onset of weakness or BOTH
Fevers, dysphagia, arthritis, muscle tenderness and fatigue are also common
UV light exposure with generalized erythema

Shawl sign: If over chest/neck

Heliotrope rash: blue-violet discoloration of eyelids, erythema crosses nasolabial folds

Gottron papules: bright pink, pale, shiny, thickened atrophic plaques over hand joints. Rash can be mistaken for eczema or psoriasis

Mechanic's hands: thickened erythematous scaly rash over palms and soles



JDM

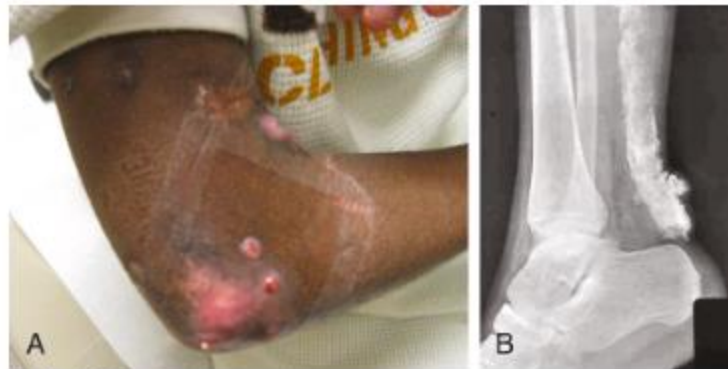
Weakness is symmetric, muscles such as neck flexors, shoulder girdle, hip flexors. Trouble combing hair, climbing stairs, getting out of bed

Esophageal and resp muscles can also be affected. Swallow test maybe helpful. Respiratory failure – hypercarbia with hypoxia, no iWOB, often need NIF

EMG shows signs of myopathy. Labs show muscle inflammation, muscle biopsy shows necrosis, serologic testing, ANA is pos in >80% of patients, RF is neg, Anemia, ESR nml. Xrays can show calcifications

Tx: MTX, steroids, IVIG (Ped Rheum is managing), PT/OT, avoid sun

Prognosis: Mortality 1% but little known about long term consequences, respond well to therapy



Calcifications in dermatomyositis. **A**, Skin effects of calcification. **B**, Radiographic evidence of calcification.

Consensus Classification of Pediatric Behçet Disease

From Koné-Paut I, Shahram F, Darce-Bello M, et al, for PEDBD group. Consensus classification criteria for paediatric Behçet's disease from a prospective observational cohort: PEDBD. *Ann Rheum Dis* . 2016;75:958–964.

ITEM	DESCRIPTION
Recurrent oral aphthosis	At least three attacks/year
Genital ulceration or aphthosis	Typically with scar
Skin involvement	Necrotic folliculitis, acneiform lesions, erythema nodosum
Ocular involvement	Anterior uveitis, posterior uveitis, retinal vasculitis
Neurologic signs	With the exception of isolated headaches
Vascular signs	Venous thrombosis, arterial thrombosis, arterial aneurysm

Behcet disease (BD)

Spectrum of symptoms is broad - vasculitis of arteries + veins

Autoinflammatory nature: polygenic disorder, innate immunity activation

Prevalence in children is 10% of adult prevalence +FH in 20% of cases.

Mean age at first symptom 8-10 years of age.

Starts with painful oral ulcer does not scar, genital ulcers heal with scars.

Anterior Uveitis. Skin lesions like erythema nodosum

Meet **3 of the criteria** in chart

Tx: Azathioprine for eye disease. Topical tx for ulcers, colchicine, sucralfate, steroids. Mortality in children with BD low except if they get pulm aneurysm



FIG. 203.1

T2-weighted MRI of a child with Sjögren syndrome showing parotitis (arrows).

Rheumatol. 2016;34:343–351, Table 1.

Juvenile SS

CLINICAL SYMPTOMS

1. Recurrent parotitis or parotid enlargement
2. Recurrent conjunctivitis (nonallergic and noninfectious)
3. Recurrent vaginitis
4. Systemic: fever of unknown origin, arthralgias, hypokalemic paralysis, or abdominal pain

OBJECTIVE

5. Ocular dryness (ocular staining or a Schirmer test)
6. Abnormal sialography
7. Elevated serum amylase
8. Leukopenia or elevated ESR
9. Hyperimmunoglobulinemia (polyclonal)
10. Renal tubular acidosis

SEROLOGY

11. At least one of anti-SSA, anti-SSB, high titer ANA (speckled pattern), RF

HISTOPATHOLOGY

12. Lymphocytic infiltration of salivary glands or other organs

DIAGNOSIS OR CLASSIFICATION REQUIREMENTS

13. Diagnosis requires at least 4 of 12 items

Juvenile Sjogren's Disease

Rare in children, classic signs of dry eyes and dry mouth

Mean age at diagnosis 9-10 yr. 75% are F

Diagnostic criteria in children proposed but not validated

Recurrent **parotid gland enlargement** and **parotiditis** common in children (>70%). Children higher rate of systemic sx, fever, adenopathy compared to adults. In kids also dental caries as a result of dry mouth.

Diagnosis based on clinical features, supported by biopsy of salivary or parotid gland -> lymphocytic infiltration (children more likely to have nml salivary and abnormal parotid biopsies) need **4 of 12 items**

Elevated ESR, positive RF, hypergammaglobulinemia, presence of SSA and SSB antibodies

Treatment: Artificial tears, massage parotids, oral lozenges, fluids. Steroids, NSAIDS, hydroxychloroquine



derma. Dermal atrophy of localized scleroderma, demonstrated by "cliff" and visible veins in a teenage female with generalized morphea.



Scleroderma (systemic and localized)

Annual incidence rare in children 1-3 per 100,000 children

Mean age onset 7-9 years. More common in Females

Most cases of localized scleroderma, onset starts in childhood. Systemic only 10%

Systemic scleroderma: pulmonary fibrosis, myocardial fibrosis, myopathy, joint pain, more favorable in children than adults

Localized: usually limited to skin and subcutaneous tissue

Systemic sclerosis. Patients with calcinosis cutis, Raynaud phenomenon, esophageal dysfunction, sclerodactyly, and telangiectasia (CREST) syndrome



FIG. 204.2
Characteristic erysipeloid erythema associated with familial Mediterranean fever. This rash appears during a flare and overlies the ankle or dorsum of the foot.

Diagnostic Criteria for Familial Mediterranean
From Livneh A, Langevitz P, Zemer D, et al. *Crit*
diagnosis of familial Mediterranean fever. *Arth*
1997;40:1879–2045.

MAJOR CRITERIA

1. Typical attacks [†] with peritonitis (generalized)
2. Typical attacks with pleuritis (unilateral) or pericarditis
3. Typical attacks with monoarthritis (hip, knee, ankle)
4. Typical attacks with fever alone
5. Incomplete abdominal attack

MINOR CRITERIA

1. Incomplete attacks [†] involving chest pain
2. Incomplete attacks involving monoarthritis
3. Exertional leg pain
4. Favorable response to colchicine

Familial Mediterranean Fever (FMF)

Autosomal recessive pathogenic variant in MEFV (amino acid protein denoted pyrin (greek for fever), found in granulocytes, monocytes and in fibroblasts. Causing anto-inflammatory disease.

Occurs primarily among Mediterranean ancestry (Jewish, Turkish, Armenian, Arab, Italian)

Short Recurrent (1-3d) fever, serositis, mono or pauciarticular arthritis or rash. Most present with sx in childhood, 90% present before 20 years. So fever + abd pain or chest pain, arthritis and rash = sounds familiar.

Distinguishing factors: in between fever episodes, no symptoms and cutaneous finding erysipeloid erythematous rash on ankle.

Episodes triggered by stress, infections or menses. Inflammatory markers will be elevated.

Frequency of attacks vary from weekly to one to two flares per year

Tx: prophylactic daily oral colchicine decreases frequency, severity intensity of flares



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Summary!

I know it's confusing

Distinguishing factors:

***JIA – younger child, so many types, remember systemic type

Ankylosing Spondylitis – M, SI joint pain, restriction in ROM, HLA-B27

Psoriasis – Nail findings, rash, joint pain

Reactive Arthritis – after infection, joint pain, uveitis, fatigue, skin findings, HLA-B27

pSLE – similar to adult SLE, multiorgan involvement, complement markers decreased

Behcet – genital and oral ulcers, uveitis, vasculitis

JDM – rash + muscle weakness

Sjogren's – parotiditis, eye dryness

Scleroderma – calcinosis + fibrosis – cutaneous or systemic

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Back to the case:

3 yr old M brought into clinic by his mother for about 2 months of swelling in his b/l knees. He went to the Urgent Care 3 months ago for a URI. It stops him from running around as much. He has also felt warm for the past 2 weeks especially in the evenings and some red/pink circles on his skin during that time which go away on it's own.

On exam, his conjunctiva are pale bilaterally, there is palpable cervical, inguinal and popliteal lymphadenopathy, his liver is 3cm below the costal margin. His Vitals include HR 130 while calm.

If you got blood work on this patient, what do you think it would show ?

Diagnosis: **sJIA**

Blood work: **elevated WBC, inflammatory markers, anemia, elevated tonormal PLT**

Tx: **NSAIDs (Naproxen)**



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Sources:

Nelson's Textbook of Pediatrics

Zitelli and Davis' Atlas of Pediatric Physical Diagnosis

Clinical Overview JIA: Bayefsky, Sarah D., MD; Rutstein, Beth H., MD, MSCE

For JIA – watch Lecturio Medical

Thank you for listening!

Lecture Feedback:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>

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